

Presenter Script (Readable)

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Slide 1: Predicting Bus Delays in Luxembourg

00:00 / 00:25

Operations support • Delayed = ≥ 1 minute late

Find when and where delays are most likely, so operators can adjust service and improve reliability.

Explain: Find when and where delays are most likely, so operators can adjust service and improve reliability.

Hi, I'm Marcelo Rech. In this project, I predict whether a bus stop event will be Delayed or On-time.

I label an event as delayed when the bus is at least 1 minute late. The goal is practical: help operations teams spot high-risk situations—like certain times, routes, or conditions—so they can make small changes that improve reliability for passengers.

I'll show the data, the main patterns I found, the model results, and the key recommendations.

Slide 2: Agenda

00:00 / 00:35

Plan • What you will see • Main score: F1

A simple path: problem → data → patterns → model → what to do with it.

Explain: A simple path: problem → data → patterns → model → what to do with it.

This is the plan for the presentation. First, I explain the problem and the goal: support operations by spotting when delays are more likely.

Then I quickly describe the data and the basic checks to confirm it is reliable enough to use. After that, I show the main patterns—like which hours, routes, and conditions have more delays.

Next, I explain the model and how I evaluate it. My main score is F1 because it balances precision and recall.

If you only remember one thing from this slide, it's this: we go from the real problem to a model that leads to practical actions.

Slide 3: Why bus delays matter (and what we are trying to solve)

00:00 / 01:10

Problem • Impact • Objective

Even small delays can break connections and reduce trust. The goal is to support operations with an early warning signal.

Explain: Even small delays can break connections and reduce trust. The goal is to support operations with an early warning signal.

On this slide I explain the “why”. Bus delays are not only a number—they affect real trips.

Even a small delay can break a connection or make passengers feel the service is not reliable. For operations, the challenge is timing: it's hard to act early if you only notice delays after they spread.

So the goal of this project is simple: predict “Delayed” versus “On-time” for each stop event. I define delayed as at least one minute late.

The model gives a probability, which we can treat like a risk score. That is more useful than a strict yes/no.

For success, I use F1 as the main score because it balances two things: catching real delays and avoiding too many false alerts. I also report precision, recall, and ROC AUC for context.

The main point here is: the model should lead to practical actions, not just a metric.

Slide 4: Data source and workflow

00:00 / 00:55

Data • Where it comes from • How we prepared it

One cleaned dataset is used for both analysis and modeling, so the numbers always match.

Explain: One cleaned dataset is used for both analysis and modeling, so the numbers always match.

This slide explains where the data comes from and how the project is organized. I start with stop-event records, which contain scheduled and actual times.

From that, I calculate the delay in minutes. I also add a small amount of extra context, like weather, because conditions can affect traffic and travel time.

The most important point is consistency: I keep one cleaned dataset and use it everywhere. That way, the patterns I show in EDA are based on the same data that the model learns from.

In practice, my workflow is: extract the data and export a CSV, clean it and check quality, explore the main patterns, create features, and then train models. I evaluate with F1 as the main score, and I also show precision, recall, and ROC AUC.

The main point here is: every result in the deck comes from a clear and repeatable process.

Slide 5: Dataset schema and scale

00:00 / 00:55

Dataset overview • One row = one stop event

Each row is one bus stop event with a scheduled time and an observed time. With millions of rows, we can spot patterns and train models with confidence.

Explain: Each row is one bus stop event with a scheduled time and an observed time. With millions of rows, we can spot patterns and train models with confidence.

This is the base dataset used for the whole project. Each row represents one stop event: we have the planned time and the real observed time.

The delay in minutes is simply the difference between those two timestamps. The dataset is large — over 7 million stop events — which is great because it makes the patterns more reliable.

In terms of data quality, almost every column is complete. The only meaningful missing field is the operator name, and that happens in about 6 thousand rows, which is less than 0.1%.

I replaced those missing operator values with "Unknown" so we keep every observation. From an operations point of view, this schema is practical because it lets us slice delay risk by time, by route/stop, and by operator.

And for modeling, we simplify the problem into a clear label: delayed if it is at least 1 minute late, otherwise on-time. All records were collected between July 25 and September 16, 2025, so the analysis reflects about 7 to 8 weeks of operations.

Numbers to mention

0.1 • 7 • 6 • 25 • 16 • 2025 • 8

Slide 6: What we predict: delayed vs on-time (and how common each one is)

00:00 / 01:15

Target • Class balance • Why F1 matters

For each stop event, the model predicts one of two outcomes: Delayed or On-time .

Explain: For each stop event, the model predicts one of two outcomes: Delayed or On-time .

On this slide I lock in the exact definition of the target. For every stop event, I label it as Delayed if it is 1 minute late or more. Otherwise, it is On-time. The donut chart shows the mix in the dataset: about 64% on-time and 36% delayed. That means delays are not rare, but on-time is still the bigger group. This is important for evaluation. If I only used accuracy, a model could look “good” just by predicting on-time too often. Instead, I use F1 as the main metric because it balances two needs: catching real delays and not creating too many false alerts.

Numbers to mention

64 • 36

Slide 7: Most delays are small, but a few are very large

00:00 / 00:55

Data quality • Delay sizes • Rare extreme events

Before modeling, I check missing values and I look at the delay distribution to understand what “normal” looks like.

Explain: Before modeling, I check missing values and I look at the delay distribution to understand what “normal” looks like.

Before modeling, I do two quick checks: missing values and the size of delays. First, missing values are very small, so the dataset is reliable enough to work with. Second, I want to explain what a “normal” delay looks like. In the top chart, most delays are very close to zero. The two dashed lines are simple cutoffs: P95 = 5 minutes means 95 out of 100 delays are 5 minutes or less. P99 = 9 minutes means 99 out of 100 delays are 9 minutes or less. So the main takeaway is: almost all delays are under 10 minutes. The bottom chart shows the rare cases above 15 minutes. These are uncommon, but they matter because they usually come from disruptions.

Numbers to mention

5 • 95 • 100 • 9 • 99 • 10 • 15

Slide 8: Some hours have more delays than others

00:00 / 00:50

EDA • Time of day • When delays happen more

This chart shows the share of stop events that are delayed for each planned hour.

Explain: This chart shows the share of stop events that are delayed for each planned hour.

Here I’m asking a simple question: does delay risk change during the day? The y-axis is the percentage of stop events that are delayed, using the same rule as before: Delayed means 1 minute late or more. The dashed line is the baseline — the overall average delayed share. You can see that some hours are clearly above baseline. The chart highlights the top hours with red markers. These tend to happen around the morning commute and late afternoon. The key point is simple: instead of treating every hour the same, operations can focus monitoring and small actions during these higher-risk windows — for example, adding a bit of schedule buffer, closer dispatch attention, or faster response when delays start to grow.

Slide 9: Weekends are slightly better, and evening peak is the riskiest band

00:00 / 00:55

EDA • Weekday vs weekend • Time bands

Same metric as before: delayed share (Delayed = ≥ 1 minute). Dashed line = overall baseline.

Explain: Same metric as before: delayed share (Delayed = ≥ 1 minute). Dashed line = overall baseline.

Here I group the same delay metric in two simple ways: by weekday and by time band. On the weekday chart, the pattern is pretty stable.

Most weekdays sit close to the baseline. The weekend is a bit better, and Sunday is the lowest.

The time-band chart shows a clearer difference. Night has the lowest delayed share, while midday and especially evening peak are higher than the baseline.

The key point is simple: weekday changes are small, but time-of-day changes are larger. So operationally, it makes sense to focus extra attention during the evening peak.

And for modeling, weekday and time band are easy features that capture routine behavior.

Slide 10: Some routes have more delays than others

00:00 / 00:50

EDA • Routes • Where delays happen more

We rank routes by delayed share (Delayed = ≥ 1 minute). We only keep routes with many events so the ranking is fair.

Explain: We rank routes by delayed share (Delayed = ≥ 1 minute). We only keep routes with many events so the ranking is fair.

After looking at time patterns, I ask a new question: are some routes worse than others? These two charts rank routes by the percent of stop events that are delayed.

The dashed line is the overall baseline across the system. The top chart shows the worst routes — routes that are clearly above the baseline.

The bottom chart shows the best routes — a helpful reference for what good performance looks like. To keep this fair, I only include routes with many events.

The key point is simple: if we want quick improvements, the worst routes are a good place to start. And in the next slide, I zoom in further to see which stops are the main bottlenecks.

Slide 11: Some stops are much “more likely to be delayed” than others

00:00 / 00:50

EDA • Stop hotspots • Where delays happen most

We rank stops by delayed share (Delayed = ≥ 1 minute). We only keep stops with many events so the ranking is fair.

Explain: We rank stops by delayed share (Delayed = ≥ 1 minute). We only keep stops with many events so the ranking is fair.

Now that we understand time patterns, I zoom in on locations. This slide ranks stops by the share of events that are delayed.

The top chart shows the worst stops — the ones with delay rates well above the baseline. The bottom chart shows the best stops — a useful reference for what good looks like.

I only include stops with many events, so we don't get “top 10” results caused by a small sample. The main takeaway is simple: a small set of stops is a good place to start operational work.

These stops can be checked for typical causes like intersections, traffic signals, high boarding time, or local congestion. This creates a short list of places where small fixes can help many trips.

Numbers to mention

10

Slide 12: Delay rates vary by operator — but this is not a “scoreboard”

00:00 / 00:45

EDA • Operator view • Useful signal (with caution)

This is helpful for monitoring. But operators run different routes and areas, so we must compare carefully.

Explain: This is helpful for monitoring. But operators run different routes and areas, so we must compare carefully.

This slide looks at delays by operator. The dashed line is the overall baseline, so bars to the right of it mean “more delays than average”, and bars to the left mean “fewer delays than average”.

The important message is: this is a useful signal, but it is not a scoreboard. Operators do not all work in the same conditions. One operator might run more peak-hour routes, or serve areas with more traffic. So if an operator is above the baseline, the next step is simple: drill down to the same routes and the same time bands, and then check if the pattern still holds.

Slide 13: When delays happen, some time bands have bigger delays

00:00 / 00:45

EDA • Severity • How big delays get

Earlier we looked at how often delays happen. Here we look at how many minutes the delay is (only for delayed events).

Explain: Earlier we looked at how often delays happen. Here we look at how many minutes the delay is (only for delayed events).

So far we measured “how often” delays happen. Now I look at “how big” delays get, in minutes, and only for events that are delayed (at least 1 minute).

Each box shows the typical delay range for a time band, and the line inside is the middle value. The dots represent rare large delays.

The main idea is that some time bands don’t just have more delays — they can also have bigger delays. This helps decide where adding a little buffer or faster dispatch response could reduce the most painful cases.

The y-axis is capped so the chart stays readable.

Slide 14: Correlation heatmap: what moves together (and a quick leakage reminder)

00:00 / 00:55

EDA • Correlation • Quick overlap check

Correlation is a simple way to see which variables “travel together”. It helps spot overlap and obvious mistakes.

Explain: Correlation is a simple way to see which variables “travel together”. It helps spot overlap and obvious mistakes.

This slide is a quick overlap check. Correlation tells us which variables move together.

For example, the rain-related features are strongly related because they describe the same weather situation. Weekday and weekend also overlap because they are two views of the same calendar idea.

The target correlations are very small here, which is realistic: delays usually come from a combination of factors, not one single number. Finally, this is also a leakage reminder: we must only use information available before the bus arrives, so we never use actual arrival time as a feature.

Slide 15: Modeling setup: what we use as inputs (and how we test fairly)

00:00 / 01:10

Modeling • What the model sees • Fair evaluation

We predict Delayed vs On-time (Delayed ≥ 1 min) using only information known before the bus reaches the stop.

Explain: We predict Delayed vs On-time (Delayed ≥ 1 min) using only information known before the bus reaches the stop.

This slide is my “model contract”: what the model is allowed to use and how I evaluate it. The goal is to predict delayed versus on-time using only information we would know before the stop happens.

I group features into numeric values, simple yes/no flags, small categories like operator and time band, and large ID fields like stop and route. For the baseline Logistic Regression, I keep it simple and interpretable, so I exclude the large ID fields. For the tree model, I can include IDs with an encoding that also handles unknown values. To test fairly, I split the data into train and test and keep the same delayed share in both.

Because delayed is about 36%, accuracy can be misleading, so I use F1 as the main metric and support it with precision, recall, and ROC AUC. Finally, Leakage is when the model accidentally sees the answer.

For example, if we use the real arrival time or the delay minutes, the prediction becomes unrealistic because we would not have that information before the bus arrives.

Numbers to mention

36

Slide 16: Which model performs best for delay alerts?

00:00 / 00:55

Modeling • Compare models • Main score: F1

We compare a simple baseline to stronger tree models, and we choose the one with the best balance between missed delays and false alarms.

Explain: We compare a simple baseline to stronger tree models, and we choose the one with the best balance between missed delays and false alarms.

Here I compare different models on the same test set. Because delays are only around one third of the data, accuracy can look good even when the model is not useful, so I focus on F1.

F1 is a balance between precision and recall. Recall tells me how many real delays I catch, and precision tells me how many of my alerts are actually true delays.

In the table, the tree models perform much better than the simple Logistic Regression baseline. The Random Forest is the best overall, and the two training options (undersampling and class_weight) are quite close.

Next, I take the best model and choose a probability threshold, because operations need a clear yes/no alert.

Slide 17: Turning model probability into a clear "Delay alert"

00:00 / 01:25

Modeling • Threshold choice • Confusion matrix

The model gives a probability. We choose a threshold to decide when to flag Delayed vs On-time (Delayed ≥ 1 min).

Explain: The model gives a probability. We choose a threshold to decide when to flag Delayed vs On-time (Delayed ≥ 1 min).

The Random Forest does not output "Delayed" or "On-time" directly — it outputs a probability. This slide shows how we turn that probability into a clear alert by choosing a threshold.

In the top chart, if we lower the threshold, we catch more real delays, but we also raise more false alarms. If we raise the threshold, we reduce false alarms, but we miss more true delays.

We pick the threshold that gives the best balance using F1, which happens around 0.46 here. At this point, recall is about 0.83, meaning we catch roughly 83% of real delays, and precision is about 0.51, meaning around half of our alerts are real.

The confusion matrix at the bottom turns this into counts, which is helpful for operations. And if the business later prefers fewer false alarms or fewer missed delays, we can move the threshold up or down.

Numbers to mention

0.46 • 0.83 • 0.51 • 83

Slide 18: Do the predictions separate delayed vs on-time well?

00:00 / 01:25

Modeling • Validation • ROC & PR curves

These curves check if the model is truly useful, not just at one threshold. (Delayed = ≥ 1 minute.)

Explain: These curves check if the model is truly useful, not just at one threshold. (Delayed = ≥ 1 minute.)

This slide checks if the model is genuinely useful, not just at one chosen threshold. The ROC curve is a simple "ranking test": if I pick one delayed event and one on-time event, does the model usually give a higher score to the delayed one?

The dashed line is random guessing, and the more the curve bends toward the top-left, the better. The PR curve is more practical here because delays are not 50/50.

The dashed PR line is the baseline delay rate, so anything above it is better than guessing. PR also shows a real operational trade-off: if we push recall higher to catch more delays, precision usually drops, meaning more false alarms.

That's why we chose a threshold on the previous slide.

Numbers to mention

50

Slide 19: What information does the model rely on most?

00:00 / 01:00

Modeling • Interpretability • What the model uses most

This chart ranks features by importance inside the Random Forest. Bigger bars mean the model used that feature more.

Explain: This chart ranks features by importance inside the Random Forest. Bigger bars mean the model used that feature more.

This slide answers a simple question: what information does the model use the most to predict delays? The bars are feature importance scores from the Random Forest.

A bigger bar means that feature helped the model more during prediction. This is not "cause and effect"; it is "usefulness for prediction".

In our case, demand and location signals stand out, like stop volume and stop/route identifiers. Time features also help, because delays follow daily patterns, but they are not the strongest drivers by themselves.

Weather features show up, but they contribute less here. The practical message is: if we want to improve reliability, the first places to investigate are busy stops and specific routes where delay risk concentrates.

Slide 20: Recommendation: use the model as a “risk list” for daily operations

00:00 / 01:10

Recommendation • How to use the model

The model does not “fix” delays by itself. It helps teams focus on the trips that are more likely to run late.

Explain: The model does not “fix” delays by itself. It helps teams focus on the trips that are more likely to run late.

This slide is about how to use the model in real life. The model outputs a risk score for each stop event. A higher score means the bus is more likely to be delayed. The simplest way to use this is as a daily priority list: where should dispatch look first, and where should planning focus first. I present it as traffic lights because it is easy to understand: Green means low risk and no action. Yellow means watchlist. Red means high risk and we act early, for example by monitoring more closely or sending an earlier passenger update. The cutoffs can be tuned depending on what the operator prefers: fewer false alarms or fewer missed delays. For rollout, I start in shadow mode, then a small pilot, then scale once the alerts are trusted. The model is a risk list, but real surprises like accidents or breakdowns need real-time alerts and human judgement.

Slide 21: Limitations and next steps

00:00 / 00:55

Limitations • Risks • Next steps

The model is useful, but some delays are hard to predict without real-time information.

Explain: The model is useful, but some delays are hard to predict without real-time information.

The key message is that some delays follow patterns, but many are caused by sudden events. Because we don’t use real-time traffic, incidents, or bus telemetry, the model cannot perfectly predict surprises like accidents or breakdowns. Also, our data covers about 7 to 8 weeks, so we don’t see full seasonality. The practical fix is deployment discipline: test in time order, monitor metrics by route and time band, and retrain when patterns shift. The best improvement would be adding real-time traffic and GPS/AVL signals.

Numbers to mention

7 • 8

Potential Q&A (practice answers)

Short answers you can say confidently. You can expand with examples from the charts if needed.

Problem & goal

Q: What is the goal of the project?

A: Predict whether a bus stop event will be delayed, so operations can focus on the highest-risk cases.

Q: How do you define “Delayed”?

A: Delayed means at least 1 minute late (≥ 1 min).

Data & time coverage

Q: What time period does the dataset cover?

A: July 25, 2025 to September 16, 2025 based on collected_at (about 7–8 weeks).

Q: Why does the time window matter?

A: Because patterns can change over time. With more months or seasons, we can test stability more strongly.

Metrics & evaluation

Q: Why not use accuracy as the main metric?

A: Because delays are not 50/50. F1 balances catching delays (recall) and avoiding too many false alarms (precision).

Q: What does precision vs recall mean here?

A: Recall = how many real delays we catch. Precision = how many alerts are truly delays.

Threshold & operations

Q: What does the threshold do?

A: It turns a probability into a simple alert: above the threshold = flag as delayed.

Q: Can the threshold be changed later?

A: Yes. If the team wants fewer false alarms, raise it. If the team wants to miss fewer delays, lower it.

Limits & next steps

Q: Why can't the model predict everything?

A: Many delays come from sudden events like traffic accidents or mechanical issues, and we don't have real-time signals.

Q: What would improve the model the most?

A: Add real-time traffic/incidents and bus GPS/telemetry (speed, dwell time, headway), and evaluate across more months.